gradient descent

Numerical example

x_0		x_1	x_2	y
1	$i=1$	1	2	3
1	$i=2$	2	1	4
1	$i=3$	3	3	5

$h = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$

apply gradient descent & compute new value of θ after 2 iterations.

$n = 3$ $\quad \theta_0, \theta_1, \theta_2 = ?$ $\quad \alpha = 0.1$

$d = 2$ $\quad \underbrace{=0 \quad =0 \quad =0}_{\text{initial values}}$

$\theta^T x$ $\quad \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 3 \end{bmatrix}$

$h_\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Iteration 1

$h_{\theta(1)} = 0 + 0\times1 + 0\times2$
$= 0$

$h_{\theta(2)} = 0 + 0\times2 + 0\times1$
$= 0$

$h_{\theta(3)} = 0 + 0\times3 + 0\times3$
$= 0$

$h_\theta(x^{(1)}) - y^{(1)} = (0 - 3)\times1 = -3$

$h_\theta(x^{(2)}) - y^{(2)} = (0 - 4)\times1 = -4$

$h_\theta(x^{(3)}) - y^{(3)} = (0 - 5)\times1 = -5$

now summation $\rightarrow \therefore -3 + (-4) + (-5)$
$= -12$

$\theta_0 := 0 - 0.1(-12)$
$= \underline{1.2}$

$\theta_1 :-$ summat" part

$(h_\theta(x^{(1)}) - y^{(1)})x^{(1)} + (h_\theta(x^{(2)}) - y^{(2)})x^{(2)}$
$+ (h_\theta(x^{(3)}) - y^{(3)})x^{(3)}$

$\Rightarrow (0 - 3)1 + (0 - 4)2 + (0 - 5)3$
$= -3 - 8 - 15$
$= -26$

$\theta_1 := 0 - 0.1(-26)$
$= \underline{2.6}$

$\theta_2 :-$ summat" part

$(h_\theta(x^{(1)}) - y^{(1)})x^{(1)} + (h_\theta(x^{(2)}) - y^{(2)})x^{(2)}$
$+ (h_\theta(x^{(3)}) - y^{(3)})x^{(3)}$

$\Rightarrow (0 - 3)2 + (0 - 4)1 + (0 - 5)3$
$= -6 - 4 - 15$
$= -25$

$\theta_2 = 0 - 0.1(-25)$
$= \underline{2.5}$

$h_\theta = \boxed{\theta_0} + \boxed{\theta_1}x_1 + \boxed{\theta_2}x$
$h_\theta = \underline{1.2} + \underline{2.6}x_1 + \underline{2.5}x_2$

Page No.
Date

iteration 2

$h\theta(1) = 1.2 + 2.6 \times 1 + 2.5 \times 2$
$= 8.8$

$h\theta(2) = 1.2 + 2.6 \times 2 + 2.5 \times 1$
$= 1.2 + 5.2 + 2.5$
$= 8.9$

$h\theta(3) = 1.2 + 2.6 \times 3 + 2.5 \times 3$
$= 16.5$

$h_\theta(x^1) - y^{(1)} = 8.8 - 3 = 5.8$
$h\theta(x^2) - y^{(2)} = 8.9 - 4 = 4.9$
$h_\theta(x^3) - y^{(3)} = 16.5 - 5 = 11.5$

$\theta_0 = 1.2 - 0.1(22.2)$
$= -1.02$

$\theta_1 = 5.8 \times 1 + 4.9 \times 2 + 11.5 \times 3$
$= 50.1$

1 5.8, 4.9
2 2
3 3

$\theta_1 = 2.6 - 50.1 \times 0.1$
$= -2.41$

$\theta_2 = 5.8 \times 2 + 4.9 \times 1 + 11.5 \times 3$
$= 51$

$\theta_2 = 2.5 - 5.1 = -2.6$